

Subject Name: Logistics and Supply Chain Management

Session No: 02

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2.0	Introduction of demand management, demand planning framework and structure, time, geography, product dimension, consistency of forecasted data, supply management; inventory management; production management; transportation management; location management; sales & operations planning and execution (s&op, s&oe);
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What We Will Cover

01 Demand Management

02 Demand Planning Framework

03 Time, Geography & Product Dimensions

04 Consistency of Forecasted Data

05 Supply Management

06 Inventory Management

07 Production Management

08 Transportation Management

09 Location Management

10 S&OP and S&OE



SECTION 01

Demand Management

Understanding what demand management means and why it anchors the entire supply chain

What is Demand Management?

Demand management is the supply chain process that balances customer requirements with the capabilities of the supply chain — it plans, shapes, and coordinates all sources of demand for a product or service so operations can be organised efficiently to satisfy that demand.



EXAMPLE



A smartphone maker uses demand management to combine online pre-orders, retail sell-through data, and carrier commitments into one coordinated view of expected demand before committing production.

Objectives of Demand Management



Accurate Forecasting

Generate a realistic, unbiased view of future demand across products, channels and time.



Demand-Supply Balance

Align what customers want with what the supply chain can profitably deliver.



Cross-Functional Alignment

Create one shared number that sales, marketing, finance and operations all plan against.



Demand Shaping

Use pricing, promotions and incentives to influence demand toward available supply.



Service Level

Protect customer service and fill rates while controlling inventory investment.



Responsiveness

Detect demand signals early and adjust plans faster than the competition.

Demand Management vs. Demand Forecasting

Demand Forecasting

- A single analytical activity: predicting future demand from data
- Statistical / quantitative technique-driven
- Produces one output — the forecast number
- An input into the wider process

Demand Management

- The broader end-to-end process that owns demand
- Combines forecasting, shaping, prioritisation and communication
- Produces a consensus plan the business commits to
- Forecasting is one activity within it

Components of the Demand Management Process



1. Demand Data Collection

Historical sales, POS data, orders, and market intelligence gathered from every channel.



2. Statistical Forecasting

Time-series and causal models generate a baseline, unconstrained forecast.



3. Consensus & Collaboration

Sales, marketing and finance review and adjust the statistical baseline into one agreed number.



4. Demand Shaping

Promotions, pricing and allocation decisions actively influence demand toward supply capability.

Example: Demand Management at a Beverage Company

A national beverage company sells through supermarkets, e-commerce, and food-service. Its demand management team blends three inputs into one plan:

60%

Baseline statistical forecast from 3 years of POS history

25%

Sales-team input on retailer promotions and new listings

15%

Marketing input on summer campaign and weather-driven spikes

Result: a single consensus forecast that production, procurement and logistics all plan against — avoiding the stockouts that occurred when each function forecasted independently.

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SECTION 02

Demand Planning Framework & Structure

How organisations structure the people, process and data that turn raw demand signals into a plan

The Demand Planning Process Flow



The cycle typically repeats monthly, feeding directly into the Sales & Operations Planning (S&OP) meeting.

Organisational Structure for Demand Planning



Demand Planning Manager

Owens the end-to-end process; chairs the consensus review and reports forecast accuracy.



Demand Planner / Analyst

Builds statistical baselines, maintains forecasting models, and analyses forecast error by SKU.



Sales & Marketing Representatives

Provide market intelligence — promotions, new listings, competitor actions.



S&OP / Executive Sponsor

Resolves conflicts between demand and supply constraints at the monthly review.

Inputs to Demand Planning



Historical Sales

Cleaned, outlier-adjusted sell-in and sell-out history by SKU and location.



POS & Channel Data

Point-of-sale scan data revealing true consumer offtake, not just shipments.



Promotions Calendar

Planned price cuts, displays and campaigns that will lift demand.



Market Intelligence

Competitor launches, economic indicators, and category trends.



Order Book

Confirmed customer orders and contractual commitments.



New Product Plans

Launch timing and cannibalisation assumptions for new SKUs.

Outputs of Demand Planning

Statistical Forecast

- Unconstrained, model-generated baseline
- Produced by time-series / causal methods
- Ignores commercial judgement and known events
- Serves as the analytical starting point

Consensus Forecast

- Statistical baseline adjusted by business input
- Single number owned by the whole organisation
- Feeds S&OP, procurement and production plans
- Tracked for accuracy (MAPE, bias) every cycle

Example: Demand Planning Framework at an FMCG Company

A packaged-foods company runs a monthly demand planning cycle with a dedicated team of 6 planners covering 40 categories.

- Week 1: Planners pull POS and shipment history; statistical baseline generated in the forecasting system
- Week 2: Sales managers review the baseline by key account and flag upcoming promotions
- Week 3: Marketing adds new-product and campaign assumptions; finance checks revenue implications
- Week 4: Consensus forecast signed off at the demand review, then handed to the S&OP meeting

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OUTCOME



Forecast accuracy (MAPE) improved from 38% to 22% within two cycles once sales and marketing began collaborating on a single number instead of forecasting separately.



SECTION 03

Time, Geography & Product Dimensions

Every forecast lives on three axes at once — when, where, and what — and must stay consistent across all three

The Time Dimension

Forecasts are built across nested time horizons, each serving a different planning decision:



Short-Term (Weekly)

Drives replenishment, scheduling and S&OE execution decisions.



Medium-Term (Monthly)

Feeds S&OP, procurement commitments and workforce planning.



Long-Term (Quarterly/Annual)

Supports capacity investment, budgeting and strategic sourcing.

Example: Time-Bucketed Forecasting

An air-conditioner manufacturer forecasts in three time buckets simultaneously:

Weekly

Buckets drive factory scheduling during the peak summer season

Monthly

Buckets set procurement of compressors and sheet metal 3 months out

Annual

Buckets justify a new production line investment for next year

Time-phasing means the weekly total for a month must reconcile exactly to that month's monthly forecast — a key consistency check covered later in this unit.

The Geography Dimension

Demand is also organised in a location hierarchy, from the broadest market down to the individual point of sale:



Example: Aggregation vs. Disaggregation

Top-Down (Aggregate first)

- Forecast total national demand for bottled water
- Split down to regions using historical share %
- Fast, stable, but misses local nuances
- Good for new products with thin store-level history

Bottom-Up (Disaggregate first)

- Forecast demand at each retail store individually
- Aggregate upward to region and national totals
- Captures local promotions and weather effects
- Needs clean, granular POS data to work well

The Product Dimension

Products are forecast at multiple levels of a hierarchy, and each level answers a different business question:



Category

e.g. “Beverages” — used for category-level budgeting and shelf-space planning.



Brand / Sub-Brand

e.g. “Sparkling Water” — used for marketing investment decisions.



SKU (Stock-Keeping Unit)

e.g. “500ml Lemon, pack of 6” — the level at which orders and production actually occur.

Example: New Product Introduction Forecasting

A snack-food company is launching a new flavour with no sales history of its own.

- Step 1: Identify a comparable "like-item" — the most recent flavour launched in the same category
- Step 2: Borrow that item's adoption curve and scale it by planned distribution and marketing spend
- Step 3: Forecast at brand level first, then allocate down to the new SKU using planned pack-mix ratios
- Step 4: Replace the borrowed curve with actual POS data after 8-12 weeks of real sales

WHY IT MATTERS



Product-dimension forecasting for new items relies on analogy and judgement, since statistical models need history that does not yet exist.

Integrating Time, Geography & Product

A demand plan is really a three-dimensional cube: every forecast cell is defined by a time bucket, a location, and a product. Planners must be able to slice the cube any way a decision requires — and every slice must reconcile to the same totals.



EXAMPLE



A retailer's monthly national forecast for "500ml water" must equal the sum of every store's weekly forecasts for that SKU rolled up across the month — this is the consistency check explored next.



SECTION 04

Consistency of Forecasted Data

Ensuring the same demand story holds true whichever level, location or time bucket you look at

04

Why Consistency Matters



Conflicting Plans

If regional totals do not sum to the national forecast, procurement and production plan against different numbers.



Inventory Imbalance

Inconsistent forecasts create stockouts in one location and excess stock in another.



Trust in the Process

Sales and finance lose confidence in a plan that changes depending on how it is sliced.



One Version of the Truth

Consistency lets every function — sales, ops, finance — work from the same committed numbers.

Methods to Ensure Consistency



Top-Down Reconciliation

Aggregate forecast is split to lower levels using fixed historical proportions.



Bottom-Up Reconciliation

Detailed forecasts are summed upward; totals become the aggregate forecast automatically.



Middle-Out Reconciliation

Forecast starts at a stable mid-level (e.g. brand) and is reconciled both up and down.

Whichever method is chosen, forecasting systems apply proportional adjustment factors so every level of the hierarchy always sums correctly — this is the core of forecast consistency.

Example: Reconciling SKU-Level and Category Forecasts

A retailer forecasts "Breakfast Cereals" at 100,000 units nationally for the month, but the sum of all individual SKU-level, store-level forecasts adds up to only 91,000 units.

$$\text{Scaling Factor} = 100,000 \div 91,000 = 1.099$$

Each SKU-store forecast is multiplied by 1.099 to reconcile to the category total

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RESOLUTION

Every detailed forecast cell is scaled up proportionally by the factor 1.099, so category, brand and SKU totals all agree — avoiding a mismatch between the demand plan used by finance and the one used by the warehouse.





SECTION 05

Supply Management

Turning the demand plan into a reliable, cost-effective flow of materials from suppliers

What is Supply Management?

Supply management is the process of identifying, evaluating and managing the suppliers and materials required to meet the demand plan — covering sourcing strategy, supplier relationships, contracts and supply risk.

Important slide



SCOPE



Supply management spans strategic sourcing (which suppliers, which contracts), tactical buying (purchase orders, delivery schedules) and supplier performance management.

Supplier Selection & Evaluation Criteria



Cost

Unit price, payment terms and total landed cost, not just quoted price.



Quality

Defect rates, certifications and consistency of incoming materials.



Delivery Reliability

On-time-in-full (OTIF) performance and lead-time consistency.



Capacity

Ability to scale volume up or down as demand plans change.



Financial Stability

Supplier's ability to survive disruption without failing to supply.



Innovation

Willingness to co-develop new materials, packaging or processes.

Managing Supply Risk



Single-Source Risk

Relying on one supplier or one region concentrates disruption risk — mitigated by dual-sourcing.



Geopolitical & Logistics Risk

Port closures, tariffs and trade restrictions can halt inbound material flow overnight.



Buffer Strategies

Safety stock of critical materials and pre-qualified backup suppliers reduce exposure.



Continuous Monitoring

Supplier scorecards and early-warning signals (financial, capacity) flag risk before it disrupts supply.

Example: Supply Management at an Automotive OEM

A car manufacturer sources semiconductors from a single overseas supplier. A regional chip shortage threatens three months of vehicle production.

- Response 1: Qualify a second chip supplier in a different region within 6 weeks
- Response 2: Redesign selected vehicle trims to use more widely available chip variants
- Response 3: Prioritise allocation of scarce chips to the highest-margin, highest-demand models

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LESSON



Strong supply management turns a potential production shutdown into a manageable trade-off — by having supplier diversity and flexible product design decided in advance, not during the crisis.



SECTION 06

Inventory Management

Holding the right stock, in the right place, at the lowest total cost

Types of Inventory



Raw Materials

Purchased inputs awaiting conversion into finished product.



Work-in-Process (WIP)

Partially completed goods still moving through production.



Finished Goods

Completed products ready for sale and distribution.



Safety Stock

Buffer inventory held to absorb demand and lead-time variability.

Inventory Costs



Holding (Carrying) Cost

Storage, insurance, obsolescence and the cost of capital tied up in stock — typically 15–25% of inventory value per year.



Ordering Cost

Administrative and processing cost incurred every time an order is placed with a supplier.



Stockout Cost

Lost sales, expediting fees and customer goodwill lost when inventory runs out.

The Economic Order Quantity (EOQ) Model

EOQ finds the order quantity that minimises the sum of ordering cost and holding cost.

$$EOQ = \sqrt{(2DS / H)}$$

D = annual demand, S = ordering cost per order, H = holding cost per unit per year

Where the curve for total ordering cost falls as order size grows, and total holding cost rises — EOQ sits at the point where the two curves cross and total cost is lowest.

Safety Stock & Reorder Point

$$\text{ROP} = (\text{Average Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$$

The inventory level that triggers a new replenishment order

Safety stock absorbs uncertainty in demand and lead time. A common approach sets it using a service-level factor (Z), the standard deviation of demand, and lead time:

$$\text{Safety Stock} = Z \times \sigma(\text{demand}) \times \sqrt{\text{Lead Time}}$$

Higher desired service level $\rightarrow$ higher Z $\rightarrow$ more safety stock

Example: EOQ & Reorder Point Calculation

A distributor sells 7,200 units/year of an item. Ordering cost = ₹500/order, holding cost = ₹20/unit/year, lead time = 10 days, average daily demand = 20 units.

$$\text{EOQ} = \sqrt{(2 \times 7,200 \times 500 / 20)} = 600 \text{ units}$$

Optimal order quantity per replenishment

$$\text{ROP} = 20 \times 10 = 200 \text{ units}$$

Reorder when stock falls to 200 units (ignoring safety stock)

INTERPRETATION

The distributor places roughly 12 orders a year ($7,200 \div 600$), each of 600 units, and triggers the next order whenever on-hand stock reaches 200 units — balancing ordering effort against holding cost.





SECTION 07

Production Management

Converting the demand plan into an executable manufacturing schedule

The Production Planning Process



Production Strategies

Important slide



Make-to-Stock (MTS)

Produced ahead of demand and held as finished-goods inventory. Fast delivery, higher inventory risk.



Make-to-Order (MTO)

Production starts only after a confirmed customer order. Low inventory, longer lead time.



Assemble-to-Order (ATO)

Standard components are stocked; final assembly happens after order, balancing speed and flexibility.

Capacity Planning & Utilisation

$$\text{Capacity Utilisation} = (\text{Actual Output} / \text{Design Capacity}) \times 100$$

Measures how much of the available capacity is actually being used



Rough-Cut Capacity Planning

Quick check of whether the master schedule is feasible against key bottleneck resources.



Detailed Capacity Planning

Machine and labour-hour level check once the schedule is confirmed.



Managing Constraints

Overtime, subcontracting or shift changes are used to close capacity gaps.

Example: Production Scheduling at a Textile Plant

A textile plant must fulfil three orders this week: 5,000m of cotton fabric (due Friday), 3,000m of blended fabric (due Wednesday), and a rush order of 1,000m denim (due Tuesday).

- Shortest-due-date rule: denim rush order is scheduled first despite being the smallest run
- Blended fabric is sequenced next to meet its Wednesday deadline
- Cotton fabric, with the most slack, fills remaining capacity through to Friday

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OUTCOME



Sequencing by due date rather than order size kept all three deliveries on time and avoided a late-shipment penalty on the rush denim order.



SECTION 08

Transportation Management

Moving materials and finished goods efficiently across the supply chain network

Modes of Transportation & Selection Criteria



Road

Flexible, door-to-door, best for short-to-medium distance and small shipments.



Rail

Low cost per unit for bulk, long-distance inland freight; slower and less flexible.



Sea

Lowest cost for international bulk cargo; long transit times.



Air

Fastest and most expensive; used for urgent or high-value shipments.

Mode selection trades off cost, speed, reliability and shipment characteristics (weight, volume, value, urgency).

Transportation Management System (TMS) Functions



Route Planning & Optimisation

Determines the most efficient sequence and mode for each shipment.



Freight Rating & Booking

Compares carrier rates and books capacity automatically.



Shipment Tracking

Provides real-time visibility of in-transit goods to planners and customers.



Carrier Performance Management

Tracks on-time delivery and damage rates to guide future carrier selection.

Important slide

Freight Cost Optimisation



Load Consolidation

Combining smaller shipments into full truckloads to cut cost per unit.



Route Optimisation

Minimising total distance and empty backhauls across delivery routes.



Carrier Negotiation

Volume commitments and multi-year contracts in exchange for lower rates.

Example: Transportation Network for an E-Commerce Company

An online retailer ships from 3 regional warehouses to customers nationwide, using a tiered transportation strategy:

Line-Haul

Full truckloads move bulk inventory between warehouses overnight by road

Last-Mile

Local courier partners handle final delivery to the customer's door

Express

Air freight is reserved only for guaranteed next-day premium orders

By matching transportation mode to delivery-speed promise rather than using one mode for everything, the retailer cut average freight cost per order by 18%.

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SECTION 09

Location Management

Deciding where facilities should sit within the supply chain network

Facility Location Decision Factors



Proximity to Demand

Closer facilities reduce delivery time and last-mile transportation cost.



Access to Transport Infrastructure

Highways, ports, rail links and airports affect inbound and outbound flow.



Labour Availability & Cost

Skilled workforce supply and wage levels shape operating cost.



Land, Tax & Regulatory Costs

Real-estate cost, incentives, and local regulations affect total cost of ownership.



Risk Exposure

Natural disaster risk and geopolitical stability of the region.



Scalability

Room to expand the facility as demand grows over time.

Warehouse / Distribution Centre Network Design

Centralised Network

- Few large warehouses serving wide areas
- Lower inventory holding cost (pooling effect)
- Longer, costlier last-mile delivery
- Simpler to manage and control

Decentralised Network

- Many smaller warehouses close to demand
- Faster, cheaper last-mile delivery
- Higher total inventory investment
- More complex to coordinate and replenish

Location-Allocation Models: An Overview

Location-allocation models mathematically decide both where to place facilities and which customers each facility should serve, typically minimising total transportation and fixed facility cost.



Centre-of-Gravity Method

Weights customer locations by shipment volume to find the lowest-transport-cost site.



Mixed-Integer Programming

Optimisation models that choose facility count, sites and customer allocation simultaneously.



Simulation Models

Test network scenarios against demand variability and service-level targets.

Example: Locating a Regional Distribution Centre

A consumer goods company needs a new DC to serve its fast-growing southern market of five major cities.

- Centre-of-gravity analysis, weighted by each city's order volume, points to a site near the region's midpoint
- The candidate site is checked against highway and rail access, land cost, and labour availability
- A nearby city with slightly higher land cost but direct highway access is chosen over the pure centre-of-gravity point

OUTCOME



The new DC cut average delivery time to the five cities from 2 days to same-day/next-day, while keeping inbound transportation cost from the factory nearly unchanged.



SECTION 10

S&OP and S&OE

The integrating processes that align demand, supply, and financial plans — monthly and week-to-week

What is Sales & Operations Planning (S&OP)?

S&OP is a monthly, cross-functional decision-making process that aligns the demand plan, the supply plan, new-product plans and financial targets into one integrated business plan endorsed by leadership.



PURPOSE



S&OP exists to reconcile what sales wants to sell, what operations can supply, and what finance can afford — into a single, agreed monthly plan.

The Monthly S&OP Cycle

1

1. Data Gathering

Collect sales, forecast and performance data

2

2. Demand Review

Sales & marketing agree the demand plan

3

3. Supply Review

Operations checks capacity and constraints

4

4. Pre-S&OP

Reconcile gaps; prepare scenarios & trade-offs

5

5. Executive S&OP

Leadership approves the final integrated plan

Sales & Operations Execution (S&OE)

S&OE is the short-term, weekly counterpart to S&OP — it manages execution against the approved plan within the current month, resolving day-to-day exceptions before they become monthly-scale problems.



Exception Monitoring

Flags material shortages, machine downtime, or demand spikes as they happen.



Rapid Resolution

Empowered weekly meetings make fast trade-off decisions within plan boundaries.



Feedback to S&OP

Recurring exceptions are escalated to the next monthly S&OP cycle for a structural fix.

S&OP vs. S&OE: A Comparison

S&OP

- Monthly cadence, 1–24 month planning horizon
- Strategic and tactical trade-off decisions
- Led by senior/executive leadership
- Sets the plan for volume, capacity and inventory

S&OE

- Weekly cadence, current-month horizon
- Operational, execution-level decisions
- Led by planning and operations managers
- Keeps actual performance aligned to the plan

Benefits of Integrated S&OP / S&OE



Better Forecast Accuracy

One consensus number replaces multiple conflicting departmental forecasts.



Improved Service & Inventory

Demand and supply stay balanced, lowering both stockouts and excess stock.



Cross-Functional Alignment

Sales, operations and finance work from the same numbers and priorities.



Faster Response to Change

S&OE resolves near-term exceptions before they disrupt the monthly plan.



Financial Visibility

The operating plan is directly tied to the revenue and margin plan.



Single Accountable Plan

Leadership formally owns one integrated plan rather than several silos.

Example: S&OP Cycle at a Consumer Electronics Company

A headphone brand runs monthly S&OP ahead of its holiday season launch.

- Demand review: Marketing raises the forecast 30% for a planned festive campaign
- Supply review: Operations confirms the factory can only flex output by 15% at short notice
- Pre-S&OP: Planners model two scenarios — air-freight expedite vs. a phased regional launch
- Executive S&OP: Leadership approves a phased launch, prioritising the top two markets first

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FOLLOW-THROUGH



Weekly S&OE meetings then tracked factory output against the approved ramp-up, catching a component delay early enough to keep the top-priority market on schedule.



KEY TAKEAWAYS

From Demand Signal to Delivered Order

- Demand management turns raw signals into one consensus plan across time, geography and product
- Supply, inventory, production, transportation and location decisions all execute against that plan
- S&OP aligns the business monthly; S&OE keeps execution on track week to week

Thank You